STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 8: Mechanical Properties of Solids

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Restoring force per unit area.

Pa.

DeltaL/L.

Within proportional limit, stress is proportional to strain.

Bulk modulus.

Yes.

Material is relatively stiff.

F/A.

5x10^5 Pa.

5x10^-4.

2x10^11 Pa.

0.01.

1x10^-4 m.

Steel carries tensile stress effectively; concrete is strong in compression.

0.001 m.

2x10^11 Pa.

3x10^5 Pa.

DeltaL proportional to L.

Proportional limit ends linearity; elastic limit is the greatest stress for full recovery after unloading.

DeltaL=FL/(AY)=3.75x10^-4 m.